Fiscal Year:	FY 2016	**Task Last Updated:**	FY 11/21/2014
PI Name:	Wotring, Virginia Ph.D.		
Project Title:	Inflight Pharmacokinetic and Pharmacodynamic Responses to Medications Commonly Used in Spaceflight		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	77030	**Congressional District:**	9
Comments:	PI formerly with Universities Space Research Association until fall 2015.		
Project Type:	Flight	**Solicitation / Funding Source:**	2013 HERO NNJ13ZSA002N-Crew Health (FLAGSHIP & NSBRI)
Start Date:	10/01/2015	**End Date:**	03/16/2016
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Norsk, Peter	**Contact Phone:**	
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Flight Program:	ISS		
Flight Assignment:	NOTE: Ended early 3/16/2016 with PI transfer to Baylor College of Medicine, per HRP (Ed., 7/28/16) NOTE: Change in period of performance due to delays per JSC HRP information; original period of performance was 12/1/2014-9/30/2020 (Ed., 6/15/16)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Barger, Laura Ph.D. (Brigham And Women's Hospital) Derendorf, Hartmut Ph.D. (University of Florida)		
Grant/Contract No.:	Internal Project		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	EDITOR'S NOTE: End date is now 3/16/2016 due to PI move to Baylor College of Medicine; continued by "Inflight Pharmacokinetic and Pharmacodynamic Responses to Medications Commonly Used in Spaceflight--NNX16AG97G" with the same investigator, Dr. Virginia Wotring. NOTE: Change in period of performance due to delays, per Johnson Space Center Human Research Program information; original period of performance was 12/1/2014-9/30/2020 (Ed., 6/15/16) Crewmembers on spaceflight missions use medications to treat spaceflight-associated conditions like space adaptation syndrome as well as common ordinary complaints. They do so under the remote guidance of their flight surgeons, who base prescriptions on their knowledge and experience in terrestrial medicine. However, even after decades of spaceflight missions, we do not know if medications act the same in the spaceflight environment as they do on Earth. Aspects of the spaceflight environment (low gravity, radiation exposure, closed environment, stress) have been shown to alter human physiology. From first principles, some of these physiological changes could be expected to alter either pharmacokinetics (PK, how the body absorbs, distributes, metabolizes, and excretes administered medications) or pharmacodynamics (PD, receptors or signalling systems that are the targets of medication action). Anecdotal data has suggested that, at least for certain medications or indications, inflight medication efficacy is poor, and spaceflight crew use high doses, repeated doses, or even experience apparent treatment failures; fortunately none of these episodes has been serious enough to cause injury or evacuation from low-Earth orbit (LEO). However, in order to prepare for exploration missions where speedy evacuation to Earth may not be a possibility, the likelihood of unexpected medication action must be determined. New studies have yielded the first documented evidence of apparent treatment failures in low Earth orbit. These instances could be explained by alterations in PK or PD. The central hypothesis of this proposal is that the spaceflight environment alters human physiology in ways that affect how medications ultimately exert their actions, either by changing PK or PD. Specifically, we propose to: (1) test the hypothesis that PK following administration of a medication is different during spaceflight as compared to ground-based data from the same subject, and (2) test the hypothesis that medication action (PD) is different during spaceflight as compared to ground-based data from the same subject. We propose to test our hypothesis with three unrelated medications commonly used on spaceflight missions: a nonbenzodiazepine hypnotic, a stimulant, and an antihistamine with antinausea action. These medications all have targets in the central nervous system, and share action on sleep or wakefulness, permitting use of the same suite of PD measures. We propose to measure PK for 3 medications frequently used by crewmembers in two different environments: on Earth and during a spaceflight mission. Circulating concentrations in the bloodstream will be measured over time after medication administration, permitting modeled estimation of absorption rates and metabolism/excretion rates. Same-subject comparisons using pre-flight and post-flight test sessions will permit measurement of same-subject variability that will provide a baseline for determination of clinically relevant changes during spaceflight. Spaceflight vs. ground differences in medication-associated effects on sleep, wakefulness, and several measures of cognitive function will be used as measures of pharmcodynamics. This study will determine if the LEO environment of the ISS alters the actions of three common medications. A likely outcome, relevant for NASA missions, is optimized dosing recommendations for one or more of these medications. In more general terms, the results of this study will indicate the likelihood and magnitude of any spaceflight-associated PK or PD changes, and thus help to determine future research priorities. The general population, who also use these medications, are also likely to benefit from additional understanding of how these medications interact with the human body.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The results of this study will indicate the likelihood and magnitude of any spaceflight-associated PK or PD changes, and thus help to determine future research priorities. The general population, who also use these medications, are also likely benefit from additional understanding of how these medications interact with the human body.
Task Progress:	New project for FY2016. EDITOR'S NOTE: End date is now 3/16/2016 due to PI move to Baylor College of Medicine; continued by "Inflight Pharmacokinetic and Pharmacodynamic Responses to Medications Commonly Used in Spaceflight--NNX16AG97G" with the same investigator, Dr. Virginia Wotring. EDITOR'S NOTE: Change in period of performance due to delays per JSC HRP information; original period of performance was 12/1/2014-9/30/2020. Period of performance is now 10/1/2015-9/30/2018 (Ed., 6/15/16)
Bibliography Type:	Description: (Last Updated: 12/24/2019)